Multi-source Distributional Test-then-pool: Risk-weighted Borrowing, Sourcewise Equivalence Fusion and Selection-aware Permutation Inference

Builds on	L. Yang, X. Liu and R. J. Evans (2026), *Data Fusion with Distributional Equivalence Test-then-pool*, arXiv:2603.11867v2.
Question	How can several historical control sources be used to test $H_0 : Q_c = Q_t$ when the sources may differ from the current control and from each other?
Content	Review of the original guarantees; two borrowing rules; exact selection-aware calibration; theorems with proofs; Monte Carlo study ($\approx 85{,}000$ synthetic data sets) with an independent-seed confirmation.
Status	Research draft, 18 October 2026. Proofs are the author's own and have not been independently refereed. Prepared with AI assistance (Claude); an earlier AI draft (GPT, 17 October 2026) is credited where its ideas are reused.
Companion	Study B: *Shrinking equivalence radii and vanishing borrowing*.
Materials	Code, seeds, raw results: `https://github.com/Yu-Jusang/ttp-extensions`

Abstract

Borrowing historical controls can raise the power of a randomized comparison with a small concurrent control arm, but heterogeneous sources can bias the comparison. We extend the distributional equivalence test-then-pool (TTP) procedure of Yang et al. (2026) from one to J historical sources, keeping the final null hypothesis $H_0 : Q_c = Q_t$ unchanged. Two borrowing rules are studied: (i) nonnegative weights chosen by minimising an *unbiased* estimate of the kernel-mean-embedding risk, built from a Gram matrix of estimated source shifts; and (ii) sourcewise MMD equivalence fusion followed by sample-size pooling of accepted sources. Both rules are data-dependent, so the final test is calibrated by *full reselection*: the entire selection is recomputed in every permutation of the current labels. We prove finite-sample validity of this calibration under explicit symmetry conditions on auxiliary bootstrap randomness, show that freezing the observed selection can only be bounded by $|\mathcal{D}|\alpha$ in general, and prove that ordinary permutation of historical data is invalid under heterogeneity. For the risk-weighted rule we prove a deterministic bound $D^2(\hat{Q}_f, Q_c^m) \le 2\hat{s}\,\hat{v}_c/m$ in every permutation, which yields an explicit Type-II error bound and consistency against all alternatives with $\sqrt{n}\,D(Q_c, Q_t) \to \infty$, uniformly over the historical distributions. Sourcewise fusion with a fixed radius admits a fixed alternative with power tending to zero. In simulations, full reselection held the nominal level in every setting, freezing the selection did not, risk weighting gained power when sources were aligned and lost power when source shifts mimicked the treatment effect, and a minP safeguard limited those losses. We do not claim finite-sample power dominance over current-only testing.

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1 Introduction

A randomized comparison of a treatment distribution Q_t with a concurrent control distribution Q_c often has far fewer control than treated units. Control outcomes from earlier studies are frequently available and can be borrowed to increase power (Viele et al., 2014; Schmidli et al., 2014). If the historical populations differ from the current one, borrowing biases the comparison, inflates Type-I error, or masks a genuine effect (Viele et al., 2014). Test-then-pool (TTP) procedures decide whether to borrow through a preliminary test. Li et al. (2020) replaced the classical equality pretest with an equivalence pretest, and Yang et al. (2026) lifted equivalence TTP to whole distributions using the maximum mean discrepancy (MMD), together with a partial bootstrap and a partial permutation that calibrate the causal test when the borrowed control is close to, but not identical with, the current control.

Yang et al. (2026) study one historical source and mention multi-source weighting only as a future direction (their Section 6). In practice several historical trials, registries or sites are available, and they differ in size, in how far they are from the current population and in *direction*: two sources can be shifted in opposite directions and partly cancel, or in the same direction as the treatment effect and partly mask it. This paper studies how to borrow from J sources while testing exactly the same null hypothesis,

$$H_0 : Q_c = Q_t \quad \text{versus} \quad H_1 : Q_c \neq Q_t,$$

and never the surrogate hypothesis $Q_f = Q_t$ for a fused control Q_f.

Contributions. We separate results taken from earlier work from new ones.

1. *Review.* We restate the original guarantees with their theorem numbers and assumptions (Section 2.2) and identify two steps whose arguments we could not verify as written: the conditional partial-permutation validity (original Theorem A.4) and the boundary behaviour of the fusion test. We do not claim that the original conclusions are false.

2. *Calibration after selection* (Section 3.2). Full-reselection partial permutation has finite-sample level $\lfloor \alpha L \rfloor / L$ for *any* symmetric borrowing rule, any historical distributions and any number B of random permutations (Theorem 3.2). We make explicit a symmetry condition on shared bootstrap randomness that is easy to violate in code. Freezing the observed selection is controlled in general only by a union bound $|\mathcal{D}|\alpha$ (Proposition 3.3), and permuting historical observations together with current ones has asymptotic size one under heterogeneity (Proposition 3.4). Full reselection was proposed in the earlier AI draft; the symmetry condition, the union bound and the invalidity result are new here.

3. *Risk-weighted borrowing* (Section 3.3). We give the fixed-weight risk decomposition of the fused kernel mean embedding, an unbiased estimate $\hat{R}_U(w) = w^\top \hat{G} w + (1-2s)\hat{v}_c/m$ of that risk, its connection to Stein-type kernel mean shrinkage (Muandet et al., 2016), and a deterministic displacement bound for the data-selected weight (Theorem 3.6). The earlier draft proved a similar bound for a conservative objective that double-counts sampling variance; the unbiased objective, its bound and the risk-safety bound (Proposition 3.7) are new.

4. *Sourcewise fusion* (Section 3.4). We specify hypotheses, statistics and bootstrap computations for J separate equivalence tests, separate the familywise error of falsely accepting a source from the final Type-I error (Proposition 3.8), and prove that a fixed radius can leave a fixed alternative with power tending to zero even under full reselection (Theorem 3.15).

5. *Power* (Section 3.5). An explicit Type-II bound for any rule with deterministic displacement control (Theorem 3.12; it sharpens a bound in the earlier draft) gives uniform $\sqrt{n}$-rate consistency of the risk-weighted test over all historical distributions (Corollary 3.13). A minP safeguard inherits the power of the current-only test at level $\alpha/2$ (Proposition 3.16).
6. *Experiments* (Sections 4 and 5). A reproduction of the original Table 2, a 5×5 factorial design with three sources, a sample-size study including a blind spot, and an independent-seed confirmation, with Monte Carlo intervals and computation times.

Consistency of a test and power gain over current-only testing at a fixed sample size are different properties. Our theory addresses the former and gives risk bounds; we do not prove that borrowing improves power, and the experiments show settings in which it does not.

2 Background

2.1 Setting and notation

We observe mutually independent samples

$$x_1^c, \dots, x_m^c \overset{iid}{\sim} Q_c, \qquad x_1^t, \dots, x_n^t \overset{iid}{\sim} Q_t, \qquad x_1^{h_j}, \dots, x_{\ell_j}^{h_j} \overset{iid}{\sim} Q_{h_j} \ (j = 1, \dots, J),$$

with empirical measures $Q_c^m, Q_t^n, Q_{h_j}^{\ell_j}$. The current pool is $Z = (Z_1, \dots, Z_N)$, $N = m + n$, whose first m coordinates are the controls.

Assumption 2.1 (Kernel)**.** k is a bounded, measurable, characteristic kernel on the outcome space with separable RKHS $\mathcal{H}_k$ and $\kappa^2 := \sup_x k(x, x) < \infty$.

The Gaussian (RBF) kernel $k(x, y) = \exp\{-\|x - y\|^2/(2\sigma^2)\}$ satisfies Assumption 2.1 with $\kappa = 1$. Boundedness is stronger than the moment condition used by Yang et al. (2026) (their Assumption 2.2); we use it to obtain non-asymptotic bounds.

The kernel mean embedding of R is $\xi_R = \mathbb{E}_{X\sim R} k(\cdot, X) \in \mathcal{H}_k$ and we write ξ_c, ξ_t, ξ_j for $\xi_{Q_c}, \xi_{Q_t}, \xi_{Q_{h_j}}$, and $\hat{\xi}_c, \hat{\xi}_t, \hat{\xi}_j$ for the embeddings of the empirical measures.

Convention for D. Throughout, $D(P, Q) := \|\xi_P - \xi_Q\|_{\mathcal{H}_k}$ denotes the MMD itself, *not* its square; D^2 is its square. Plug-in values $D(P^a, Q^b) = \|\hat{\xi}_P - \hat{\xi}_Q\|$ are the (biased, nonnegative) V-statistic estimators used by Yang et al. (2026). All final test statistics are squared MMDs; the fusion test uses D.

Fused control. For a weight vector $w \in \mathcal{W}_J := \{w \in [0, 1]^J : s(w) := \sum_j w_j \le 1\}$ define the fused control and its embedding

$$\hat{Q}_f(w) = (1 - s)Q_c^m + \sum_{j=1}^{J} w_j Q_{h_j}^{\ell_j}, \qquad \hat{\xi}_f(w) = (1 - s)\hat{\xi}_c + \sum_j w_j \hat{\xi}_j. \tag{1}$$

Literal pooling of a set S of sources corresponds to $w_j = \ell_j/(m + \ell_S)$ for $j \in S$, $\ell_S = \sum_{j\in S} \ell_j$; it is the multi-source version of $Q_f^{m+\ell}$ in Yang et al. (2026), and its embedding is exactly the weighted sum in (1) (their Lemma A.1). Shift directions and their empirical Gram matrix are

$$b_j = \xi_j - \xi_c, \qquad \hat{G}_{ij} = \langle \hat{\xi}_i - \hat{\xi}_c, \hat{\xi}_j - \hat{\xi}_c \rangle, \qquad \|\hat{\xi}_f(w) - \hat{\xi}_c\|^2 = w^\top \hat{G} w = D^2(\hat{Q}_f(w), Q_c^m). \tag{2}$$

For a distribution R let $v_R := \mathbb{E}\|k(\cdot, X) - \xi_R\|^2 = \mathbb{E}k(X, X) - \|\xi_R\|^2 \le \kappa^2$ (the RKHS trace variance), and let $\hat{v}_c = \frac{m}{m-1}\{m^{-1}\sum_i k(x_i^c, x_i^c) - \|\hat{\xi}_c\|^2\}$ be its unbiased estimate.

2.2 The original equivalence TTP

We restate the results of Yang et al. (2026) that we build on. Statements are paraphrased with the original numbering; $\gamma = m/(m+\ell)$, $\eta = n/(m+n)$, $c_1 = m/n$, $c_2 = \ell/n$.

Fusion test (original Section 3.1, Theorem 3.1). The fusion hypotheses are $H_0^f : D(Q_c, Q_h) \ge \theta$ against $H_1^f : D(Q_c, Q_h) < \theta$ for a prespecified radius $\theta > 0$. The statistic is $\Delta_{m,\ell}^f = \theta - D(Q_c^m, Q_h^\ell)$, and H_0^f is rejected (the source is *merged*) when $\Delta_{m,\ell}^f$ exceeds the $(1-\alpha_f)$ quantile of $S_{m,\ell} = D(Q_c^m, Q_c) + D(Q_h^\ell, Q_h)$, which upper bounds Δ^f under H_0^f by the triangle inequality. The quantile is approximated by Efron's bootstrap, $D_W(Q_c^m) = \|m^{-1}\sum_i (W_i - 1)k(\cdot, x_i^c)\|$ with $W \sim \text{Multinomial}(m; 1/m, \dots, 1/m)$ and similarly for Q_h^ℓ (Liu and Gandy, 2026, Section 4.2, Lemma 7). (The original text writes Multinomial$(n; \dots)$ in both places; we use the actual sizes m and ℓ.) Theorem 3.1 states, under second-moment conditions, that the rejection probability tends to 0, to some $a \in (0, \alpha_f]$, or to 1 when $D(Q_h, Q_c)$ is $>$, $=$ or $< \theta$; it is adapted from Liu and Gandy (2026, Theorem 8).

Partial bootstrap (original Algorithm 2, Proposition 3.4, Theorems 3.5–3.6). With the merged control $Q_f^{m+\ell} = \gamma Q_c^m + (1-\gamma)Q_h^\ell$, the statistic is $\Delta = \sqrt{n}\{D^2(Q_f^{m+\ell}, Q_t^n) - D^2(Q_f^{m+\ell}, Q_c^m)\}$. Each bootstrap replicate draws m and n points from Q_c^m (pseudo-control and pseudo-treatment) and ℓ points from Q_h^ℓ. Under H_0, fixed ratios (Assumption 3.2) and $Q_c \ne Q_h$, Δ and its bootstrap version are asymptotically $N(0, 4(1 + 1/c_1)\sigma_c^2)$ (Proposition 3.4), giving asymptotic level (Theorem 3.5). When $Q_h = Q_c$ the limit is degenerate and validity is not established (discussion after Theorem 3.5). Consistency holds if and only if $Q_c = Q_h$ or $2(1-\gamma)D(Q_h, Q_c)\cos\beta < D(Q_c, Q_t)$, where $\cos\beta = \langle \xi_h - \xi_c, \xi_t - \xi_c\rangle / (\|\xi_h - \xi_c\|\|\xi_t - \xi_c\|)$ (Theorem 3.6).

Partial permutation (original Section 3.2.3, Theorems 3.8–3.9). Only the current labels are permuted; the historical sample stays in the fused control. With the statistic $T_{n,m,\ell} = D^2(Q_f^{m+\ell}, Q_t^n)$ and a Monte Carlo quantile including the observed value, the test has finite-sample level for a *fixed* merging decision and every Q_h (Theorem 3.8). Consistency holds if and only if $Q_c = Q_h$ or $2D(Q_h, Q_c)\cos\beta < \{(1-\eta)^{-1} + \eta\}D(Q_t, Q_c)$ (Theorem 3.9, as printed; Study B, Section 2.2, re-derives this condition from the original proof in a different form).

Overall procedure (Theorems 3.12, 3.16; Corollaries 3.17, 3.18; Remark 3.19). If the fusion test satisfies the trichotomy of Theorem 3.1 and the causal test has conditional asymptotic level given each fusion outcome (conditions (11)–(12)), the overall TTP has asymptotic level α (Theorem 3.12); consistency of the branches yields consistency of the whole (Theorem 3.16). Corollary 3.17 asserts level with partial bootstrap when $Q_h \ne Q_c$ and with partial permutation for all Q_h; Corollary 3.18 inherits the geometric consistency conditions above. Remark 3.19 notes the resulting lack of consistency against all alternatives and suggests a radius $\theta_n = o(1)$, leaving its analysis open. That direction is pursued in the companion Study B.

2.3 Points that require care

1. *Conditioning on the fusion event.* Conditions (11)–(12) are verified in Appendix A.2 of Yang et al. (2026). For partial permutation (their Theorem A.4), the argument uses that the merge event $A_n = \{\phi_f(Q_c^m, Q_h^\ell) = 1\}$ is measurable with respect to (Z, Q_h^ℓ) and then applies the conditional permutation bound $\mathbb{P}(T \geq \hat{c} \mid Z, Q_h^\ell) \leq \alpha$. A permutation test is valid conditionally on the *unordered* pool (the orbit of Z), not on the labelled vector, and A_n depends on which values carry control labels. We could not complete the argument as written. Conditioning on the fully labelled Z makes the observed statistic deterministic; conditioning on the orbit makes A_n non-measurable. For partial bootstrap (their Proposition A.2), tightness of $A_n - A$ is transferred to the conditional law, but the Gaussian limit of the leading term is not. Our experiments (Section 5), and the null rejection rates 0.064–0.074 in their Table 2, are consistent with a finite-sample excess from conditioning.
2. *Boundary of the fusion test.* The boundary rejection limit $a \in (0, \alpha_f]$ requires non-degeneracy; for instance, two point masses at distance exactly θ give a statistic equal to 0 and a critical value of 0, so strict rejection has probability 0. None of our results needs $a > 0$.
3. *Degeneracy of the partial bootstrap at $Q_h = Q_c$.* This matters whenever only near-identical sources are merged; Study B analyses it.

2.4 Related work

Historical and external controls. Static and dynamic borrowing are reviewed by Viele et al. (2014). Bayesian approaches discount historical data through power priors (Ibrahim and Chen, 2000), commensurate priors (Hobbs et al., 2011), robust meta-analytic-predictive priors (Schmidli et al., 2014), and multisource exchangeability models that let each of several sources be exchangeable or not (Kaizer et al., 2018); the latter is the closest Bayesian analogue of our sourcewise fusion. Frequentist test-based integration includes the equivalence TTP of Li et al. (2020), the elastic integrative analysis of Yang et al. (2023), which pools real-world data only when a test indicates compatibility and tunes the threshold to minimise mean squared error, and conformal selective borrowing (Zhu et al., 2025), which selects individual external controls and calibrates by randomization inference with finite-sample exact Type-I error. Our full-reselection calibration follows the same randomization logic at the level of whole sources and distributions.

Kernel embeddings and MMD tests. The MMD two-sample test is due to Gretton et al. (2012); the empirical embedding attains the minimax rate $n^{-1/2}$ (Tolstikhin et al., 2017); permutation MMD tests are minimax rate optimal in many settings (Kim et al., 2022); and aggregation over kernels uses the same row-wise calibration as our safeguard (Schrab et al., 2023). MMD equivalence testing is due to Liu and Gandy (2026).

Weighted combination and shrinkage. Combining an unbiased small-sample estimator with a biased larger one is a Stein-type problem; Chen et al. (2015) study it for regression, and Muandet et al. (2016) show that the empirical kernel mean is inadmissible and propose kernel mean shrinkage estimators. Our risk-weighted rule shrinks the current embedding towards historical embeddings rather than towards a fixed target.

Algorithm 1 Full-reselection partial permutation test

Require: current pool Z (m controls first), historical samples $\mathbb{H}$, borrowing rule ω, auxiliary randomness U (drawn once), B, α

Draw $A_1, \dots, A_B$ i.i.d. uniform m-subsets of $\{1, \dots, N\}$; set $A_0 = \{1, \dots, m\}$
for $b = 0, \dots, B$ **do**
  $w_b \leftarrow \omega(Q^m_{c,A_b}, \mathbb{H}, U)$ ▷ re-run every data-dependent step
  $S_b \leftarrow D^2\big((1 - s_b)Q^m_{c,A_b} + \sum_j w_{b,j} Q^{\ell_j}_{h_j},\ Q^n_{t,A_b}\big)$
end for
$p \leftarrow L^{-1}\#\{b : S_b \geq S_0\}$; reject H_0 if $p \leq \alpha$

Selection-aware randomization and multiplicity. Random-permutation tests with the observed allocation included are exact (Hemerik and Goeman, 2018; Phipson and Smyth, 2010). Post-selection inference that ignores selection is invalid in general (Leeb and Pötscher, 2005). Combining dependent statistics through the joint permutation distribution of their p-values is the minP principle of Westfall and Young (1993). Restricting a nuisance supremum to a confidence set (Berger and Boos, 1994) is an alternative route that we do not pursue. Equivalence testing theory is reviewed by Romano (2005).

3 Method

3.1 Borrowing rules and the selection problem

A *borrowing rule* maps the current control sample, the historical samples $\mathbb{H} = (Q^{\ell_1}_{h_1}, \dots, Q^{\ell_J}_{h_J})$ and auxiliary randomness U to weights $\omega(Q^m_c, \mathbb{H}, U) \in \mathcal{W}_J$. The borrowed test statistic is

$$T^{\mathrm{bor}} = D^2\big(\hat{Q}_f(\omega), Q^n_t\big) = \|(1 - s)\hat{\xi}_c + \textstyle\sum_j w_j \hat{\xi}_j - \hat{\xi}_t\|^2. \tag{3}$$

Examples are no borrowing ($\omega \equiv 0$, giving the current-only statistic $T^{\mathrm{cur}} = D^2(Q^m_c, Q^n_t)$), always pooling all sources, the risk weights of Section 3.3, and sourcewise fusion followed by pooling (Section 3.4). Except for the fixed rules, ω depends on the same current control data that enter the final test. This is the selection problem: the final test must account for how the weights were chosen.

3.2 Calibration after selection

Full reselection. Let $A_0 = \{1, \dots, m\}$ be the observed control positions and let $A_1, \dots, A_B$ be independent uniformly random m-subsets of $\{1, \dots, N\}$, drawn independently of all data; set $L = B + 1$. For an allocation A, let $Q^m_{c,A}$ and $Q^n_{t,A}$ be the empirical measures of $\{Z_i : i \in A\}$ and $\{Z_i : i \notin A\}$. In row b the *whole* rule is re-applied:

$$w_b = \omega(Q^m_{c,A_b}, \mathbb{H}, U), \qquad S_b = D^2\big((1 - s_b)Q^m_{c,A_b} + \textstyle\sum_j w_{b,j} Q^{\ell_j}_{h_j},\ Q^n_{t,A_b}\big), \tag{4}$$

and the p-value is

$$p_{\mathrm{FR}} = \frac{1}{L} \sum_{b=0}^{B} \mathbf{1}\{S_b \geq S_0\}. \tag{5}$$

Historical observations are never relabelled. Ties are counted against rejection, and $p_{\mathrm{FR}} \geq 1/L$.

Assumption 3.1 (Symmetry of the rule)**.** (i) U is independent of $(Z, \mathbb{H})$. (ii) For each $(\mathbb{H}, U)$, the weights $\omega(Q^m_{c,A}, \mathbb{H}, U)$ depend on the control sample only through its multiset of values.

Assumption 3.1(ii) holds for any rule written in terms of empirical measures. It can fail through auxiliary randomness. A natural implementation draws one matrix of multinomial bootstrap weights and attaches its ith column to the ith control *by index* in every row, and the result then depends on index order. Attaching the shared weights to the controls *sorted by value*, or drawing independent weights in every row, restores symmetry. The first version of our code used index order; we corrected it before the reported runs.

Theorem 3.2 (Finite-sample level of full reselection)**.** *Suppose $Q_c = Q_t$, Assumption 3.1 holds, and $\mathbb{H}$ is independent of Z, with arbitrary historical distributions. Then for all $m, n, \ell_1, \ldots, \ell_J$ and $B \geq 1$,*

$$\mathbb{P}(p_{\mathrm{FR}} \leq \alpha) \leq \lfloor \alpha L \rfloor / L \leq \alpha.$$

The proof (Appendix A.1) shows that, conditionally on the unordered current pool, $\mathbb{H}$ and U, the value sets of $A_0, \ldots, A_B$ are i.i.d. uniform subsets, so $S_0, \ldots, S_B$ are exchangeable. Nothing is assumed about the historical distributions, the accuracy of any fusion bootstrap, or B. The guarantee concerns the final test only. It does not say that the fusion decisions inside the rows are calibrated equivalence tests.

Freezing the selection. The original procedure selects once, using the observed allocation, and keeps that selection in all resamples. With a finite set $\mathcal{D}$ of possible selections (for sourcewise fusion $|\mathcal{D}| \leq 2^J$) and p_d the fixed-rule partial-permutation p-value for selection d, the frozen test rejects when $p_{\hat{d}} \leq \alpha$ with $\hat{d} = \hat{d}(Q_c^m, \mathbb{H})$.

Proposition 3.3 (What freezing guarantees)**.** *Under H_0, $\mathbb{P}(p_{\hat{d}} \leq \alpha) \leq \sum_{d \in \mathcal{D}} \mathbb{P}(p_d \leq \alpha, \hat{d} = d) \leq \min\{1, |\mathcal{D}|\alpha\}$ for all sample sizes. Running the frozen test at level $\alpha/|\mathcal{D}|$ therefore has level α.*

The bound is a union bound and is typically loose. The mechanism of excess rejection is that $\hat{d}$ and p_d are computed from the same control values. For instance, with $Q_h = Q_c$, a source is rejected for fusion exactly when $\hat{\xi}_c$ lies far from $\hat{\xi}_h \approx \xi_c$; then $\hat{\xi}_c$ also tends to lie far from $\hat{\xi}_t$, and the current-only branch rejects too often. Asymptotically this disappears when $\mathbb{P}(\hat{d} = d^*) \to 1$ for a deterministic d^*; the oracle-coupling argument in Study B makes this precise. It does not disappear when the selection probability stays strictly between 0 and 1.

Permuting historical data. One might instead pool all historical, control and treated observations and permute every label.

Proposition 3.4 (Permuting historical data is invalid under heterogeneity)**.** *Let $J = 1$, $Q_c = Q_t$, $D(Q_c, Q_h) > 0$, fixed ratios $m/n, \ell/n$, $\alpha L \geq 1$ and B fixed. The test that always merges and permutes all $m + \ell + n$ labels has rejection probability tending to one.*

The partial permutation of Yang et al. (2026) avoids this by leaving historical observations in place; full reselection inherits this feature.

3.3 A-1: risk-weighted borrowing

Fixed-weight risk. Treat $\hat{\xi}_f(w)$ as an estimator of ξ_c.

Proposition 3.5 (Fixed-weight risk decomposition)**.** *For fixed $w \in \mathcal{W}_J$, with $s = s(w)$ and $v_j = v_{Q_{h_j}}$,*

$$R(w) := \mathbb{E}\|\hat{\xi}_f(w) - \xi_c\|^2 = \Big\| \sum_j w_j b_j \Big\|^2 + (1-s)^2 \frac{v_c}{m} + \sum_j w_j^2 \frac{v_j}{\ell_j}.$$

Moreover $\mathbb{E}\hat{G}_{ij} = \langle b_i, b_j \rangle + v_c/m + \mathbf{1}\{i=j\} v_j/\ell_j$ and $\mathbb{E}\hat{v}_c = v_c$, so

$$\hat{R}_U(w) := w^\top \hat{G} w + (1-2s)\frac{\hat{v}_c}{m} \qquad \text{satisfies} \qquad \mathbb{E}\hat{R}_U(w) = R(w) \ \text{ for every fixed } w. \tag{6}$$

The bias term $\|\sum_j w_j b_j\|^2 = w^\top G w$ with $G_{ij} = \langle b_i, b_j \rangle$ depends on the angles between shift directions, not just on the distances $\|b_j\| = D(Q_c, Q_{h_j})$. Two sources with $\langle b_1, b_2 \rangle < 0$ can be combined with smaller bias than either alone, while sources with aligned shifts compound. $\hat{R}_U$ uses the full matrix $\hat{G}$. The plug-in objective of the earlier draft, $L(w) = w^\top \hat{G} w + (1-s)^2 \hat{v}_c/m + \sum_j w_j^2 \hat{v}_j/\ell_j$, has expectation $R(w) + s^2 v_c/m + \sum_j w_j^2 v_j/\ell_j$: it counts the sampling variance of the sources twice and is biased toward less borrowing. We keep it as a comparison (WC).

Stein connection. For $J = 1$ the minimiser of $\hat{R}_U$ over $w \in [0,1]$ is

$$\hat{w} = \min\Big\{1, \ \frac{\hat{v}_c}{m\, D^2(Q_c^m, Q_h^\ell)}\Big\}, \qquad \text{so} \qquad \hat{\xi}_f = \hat{\xi}_c + \hat{w}(\hat{\xi}_h - \hat{\xi}_c)$$

is a positive-part James–Stein-type shrinkage of the current embedding towards the historical one, in the spirit of the kernel mean shrinkage estimators of Muandet et al. (2016) with a data-defined target. For general J, the interior stationary point is $\hat{w} = (\hat{v}_c/m)\hat{G}^{-1}\mathbf{1}$ when $\hat{G} \succ 0$.

Selected weights. We minimise $\hat{R}_U$ over a prespecified finite set $\mathcal{W}_g \subset \mathcal{W}_J$ containing 0 (a grid with step 0.05 and $s \le s_{\max} = 0.9$ in our experiments), with ties broken towards smaller s:

$$\hat{w} = \operatorname*{arg\,min}_{w \in \mathcal{W}_g} \hat{R}_U(w).$$

The fixed-w identity (6) does *not* make $\hat{R}_U(\hat{w})$ unbiased for the risk of the selected estimator. The minimum of unbiased estimates is optimistic: $\mathbb{E}\min_w \hat{R}_U(w) \le \min_w R(w)$. What does survive selection is a deterministic bound.

Theorem 3.6 (Deterministic displacement)**.** *Let $m \ge 2$. If $\hat{w}$ satisfies $\hat{R}_U(\hat{w}) \le \hat{R}_U(0)$, which holds for any minimiser over a set containing 0, then for every data set*

$$D^2\big(\hat{Q}_f(\hat{w}), Q_c^m\big) = \hat{w}^\top \hat{G} \hat{w} \le 2\hat{s}\,\frac{\hat{v}_c}{m} \le \frac{2\kappa^2 \hat{s}}{m-1} \le \frac{2\kappa^2}{m-1}.$$

The same holds with Q_c^m replaced by Q_{c,A_b}^m in every row of Algorithm 1.

Proposition 3.7 (Risk safety of the selected estimator)**.** *Under the conditions of Theorem 3.6, for any historical distributions,*

$$\Big(\mathbb{E}\|\hat{\xi}_f(\hat{w}) - \xi_c\|^2\Big)^{1/2} \le \Big(\frac{2\kappa^2}{m-1}\Big)^{1/2} + \Big(\frac{v_c}{m}\Big)^{1/2} \le \frac{(1+\sqrt{2})\,\kappa}{\sqrt{m-1}}.$$

Thus the selected estimator is never worse than the current-only estimator by more than a constant factor ($\le (1+\sqrt{2})^2 \approx 5.8$ when $v_c = \kappa^2$), however bad the sources. This is a worst-case guarantee only, not a promise of improvement. Reducing $\mathbb{E}\|\hat{\xi}_f - \xi_c\|^2$ also does not imply higher power. Power depends on the whole null and alternative distributions of (3), and a borrowing bias aligned with $\xi_t - \xi_c$ reduces the contrast even when it lowers the embedding risk (Section 5).

Algorithm 2 Risk-weighted borrowing with full reselection (WU-R)

Require: data, grid $\mathcal{W}_g \ni 0$ ordered by s, B, α

Precompute the Gram matrix K of all $N + \sum_j \ell_j$ outcomes; $r_j(i) = \ell_j^{-1} \sum_k k(Z_i, x_k^{h_j})$; $H_{ij} = \langle \hat{\xi}_i, \hat{\xi}_j \rangle$

Draw $A_1, \dots, A_B$; $A_0 = \{1, \dots, m\}$

for $b = 0, \dots, B$ **do**

  $c_b = \mathbf{1}_{A_b}/m$, $t_b = \mathbf{1}_{A_b^c}/n$; $\langle \hat{\xi}_c, \hat{\xi}_c \rangle = c_b^\top K c_b$, $\langle \hat{\xi}_c, \hat{\xi}_j \rangle = c_b^\top r_j$, etc.

  $\hat{G}_b = H - \mathbf{1}(c_b^\top r)^\top - (c_b^\top r)\mathbf{1}^\top + c_b^\top K c_b \, \mathbf{1}\mathbf{1}^\top$; $\hat{v}_{c,b} = \frac{m}{m-1}(\overline{k(x,x)}_{A_b} - c_b^\top K c_b)$

  $w_b = \arg\min_{w \in \mathcal{W}_g} \{w^\top \hat{G}_b w + (1 - 2s(w))\hat{v}_{c,b}/m\}$ ▷ first minimiser

  $S_b = \|(1 - s_b)\hat{\xi}_{c,b} + \sum_j w_{b,j}\hat{\xi}_j - \hat{\xi}_{t,b}\|^2$ ▷ quadratic form in K

end for

$p = L^{-1}\#\{b : S_b \geq S_0\}$

3.4 A-2: sourcewise equivalence fusion followed by pooling

Fusion tests. For each source j test

$$H_{0,j}^f : D(Q_c, Q_{h_j}) \geq \theta_j \quad \text{against} \quad H_{1,j}^f : D(Q_c, Q_{h_j}) < \theta_j$$

with the statistic $\theta_j - D(Q_c^m, Q_{h_j}^{\ell_j})$ of Liu and Gandy (2026). Draw $W^{(r)} \sim \text{Mult}(m; \mathbf{1}/m)$ and $\tilde{W}_j^{(r)} \sim \text{Mult}(\ell_j; \mathbf{1}/\ell_j)$, $r = 1, \dots, B_f$, compute

$$S_j^{(r)} = \frac{1}{m}\Big\| \sum_i (W_i^{(r)} - 1)k(\cdot, x_{(i)}^c)\Big\| + \frac{1}{\ell_j}\Big\| \sum_k (\tilde{W}_{j,k}^{(r)} - 1)k(\cdot, x_k^{h_j})\Big\|,$$

where $x_{(i)}^c$ are the controls sorted by value, so the computation is symmetric (Assumption 3.1). Each norm is a quadratic form $(W - \mathbf{1})^\top K (W - \mathbf{1})$ in the corresponding Gram block. Let q_j be the $\lceil (1 - \alpha_{f,j})B_f \rceil$-th order statistic of $(S_j^{(r)})_r$. Source j is accepted if

$$D(Q_c^m, Q_{h_j}^{\ell_j}) + q_j < \theta_j. \tag{7}$$

The current-control bootstrap weights can be shared across sources. The historical weights are label-invariant and are computed once.

Pooling, empty sets and the final test. The accepted set $\hat{S}$ is pooled with sample-size weights $w_j = \ell_j \mathbf{1}\{j \in \hat{S}\}/(m + \ell_{\hat{S}})$. If $\hat{S} = \emptyset$, the rule returns $w = 0$ and the final statistic reduces to the current-only statistic, so no separate branch is needed. The final test uses the statistic T^{bor} with full reselection (SW-R): (7) is recomputed in every row with that row's control sample and the same $W^{(r)}, \tilde{W}^{(r)}$. For comparison we also run the frozen version with partial permutation (SW-F), the frozen version with the original partial bootstrap (SW-PB, with a current-only permutation test when $\hat{S} = \emptyset$), and a sample-splitting version (SW-S, Proposition 3.11).

Multiplicity: two different error rates. A *false acceptance* is the inclusion of a source with $D(Q_c, Q_{h_j}) \geq \theta_j$. Its familywise rate concerns the fusion stage only.

Proposition 3.8 (Fusion-stage familywise error)**.** *Let* $\mathcal{J}_0 = \{j : D(Q_c, Q_{h_j}) \geq \theta_j\}$. *Then* $\mathbb{P}(\hat{S} \cap \mathcal{J}_0 \neq \emptyset) \leq \sum_{j \in \mathcal{J}_0} \mathbb{P}(j \text{ accepted})$. *If each fusion test has asymptotic level* $\alpha_{f,j}$ *under* $H_{0,j}^f$ *(as*

in original Theorem 3.1), then $\limsup \mathbb{P}(\hat{S} \cap \mathcal{J}_0 \neq \emptyset) \leq \sum_j \alpha_{f,j}$*; Bonferroni levels* $\alpha_{f,j} = \alpha_f/J$ *give familywise error at most* α_f*, with no independence assumption.*

This proposition is a statement about the fusion stage only. The final Type-I error of SW-R is controlled by Theorem 3.2 whatever the fusion levels are. Conversely, a Bonferroni-adjusted fusion stage does *not* by itself make a frozen final test valid: Proposition 3.3 still applies, because the selection and the final statistic share the control data. The choice of $\alpha_{f,j}$ affects power, not the final level of SW-R.

Lemma 3.9 (Displacement after an equivalence gate)**.** *In every row,* $D(\hat{Q}_{f,b}, Q^m_{c,A_b}) \leq \rho \max_j \theta_j$ *with* $\rho = \ell_{\text{tot}}/(m + \ell_{\text{tot}})$*,* $\ell_{\text{tot}} = \sum_j \ell_j$*.*

Lemma 3.10 (Order of the fusion critical value)**.** *Let* $k_f = \lceil (1 - \alpha_{f,j}) B_f \rceil$*. For every row,* $\mathbb{E}q_j^2 \leq 2\kappa^2(1/m + 1/\ell_j)\, B_f/(B_f - k_f + 1)$*. In particular* $q_j = O_P(n^{-1/2})$ *under fixed ratios if* $B_f/(B_f - k_f + 1)$ *stays bounded.*

Sample splitting. Split the controls at random into a pilot set of size m_1 and a test set of size $m_2 = m - m_1$. Select $\hat{S}$ (or $\hat{w}$) from the pilot and the historical data, then run a fixed-rule partial permutation on the test controls and the treated units, with weights $\ell_j/(m_2 + \ell_{\hat{S}})$.

Proposition 3.11 (Sample splitting)**.** *Under* H_0 *the split test has level* $\lfloor \alpha L \rfloor / L$ *for every selection rule.*

Splitting pays for exactness with m_1 fewer controls in the final comparison, which is costly exactly when m is small.

3.5 Power

Let $d = D(Q_c, Q_t)$, $k_\alpha = \lfloor \alpha L \rfloor$, and

$$v_{m,n} = \kappa^2 \Big(\frac{1}{m} + \frac{1}{n} \Big), \qquad M_{m,n} = \frac{\kappa^2 N^2}{mn(N-1)}.$$

Both are $O(n^{-1})$ under fixed ratios.

Theorem 3.12 (Type-II error under displacement control)**.** *Suppose* $k_\alpha \geq 1$ *and, in every row of Algorithm 1,* $D(\hat{Q}_{f,b}, Q^m_{c,A_b}) \leq \delta$ *for a constant* δ*. If* $d > 2\delta$*, then*

$$\mathbb{P}(p_{\text{FR}} > \alpha) \leq \min \Big\{ 1,\ \frac{4\,(v_{m,n} + B M_{m,n}/k_\alpha)}{(d - 2\delta)^2} \Big\}.$$

Since $B/k_\alpha \leq 2/\alpha$ when $\alpha L \geq 1$, the bound is of order $\{n(d - 2\delta)^2\}^{-1}$. It is crude at small n but explicit. It does not require independence of the permutation indicators, and it holds for any distributions, including sample-size-dependent ones. With $\delta = 0$ it applies to the current-only test. The earlier draft proved a bound of the same form with a different exceedance count; the argument in Appendix A.5 uses k_α directly.

Corollary 3.13 (Uniform consistency of risk-weighted borrowing)**.** *For WU-R, Theorem 3.6 gives* $\delta = \delta_m = \kappa\sqrt{2/(m-1)}$*. Under fixed ratios and fixed B with* $\alpha L \geq 1$*,*

$$\sup \big\{ \mathbb{P}(p_{\text{FR}} > \alpha) :\ D(Q_c, Q_t) \geq d_n,\ \text{any } Q_{h_1}, \ldots, Q_{h_J} \big\} \to 0 \quad \text{whenever} \quad \sqrt{n}\, d_n \to \infty.$$

In particular WU-R is consistent against every fixed alternative.

This is the same detection rate as for current-only testing. The rate is not shown to be optimal, and the corollary does not say that WU-R is more powerful than current-only testing at any fixed n.

Corollary 3.14 (Sourcewise fusion with a fixed radius)**.** *SW-R rejects with probability tending to one whenever* $D(Q_c, Q_t) > 2\rho \max_j \theta_j$.

The condition in Corollary 3.14 is only sufficient. The next result shows that some restriction is necessary.

Theorem 3.15 (Fixed-radius blind spot)**.** *Let all sources share a distribution* Q_h *with* $0 < D(Q_c, Q_h) < \theta$ *(*$\theta_j \equiv \theta$*), let ratios* m/n *and* ℓ_j/n *be fixed, and let*

$$Q_t = \gamma Q_c + (1-\gamma)Q_h, \qquad \gamma = \frac{m}{m + \ell_{\text{tot}}}.$$

Then $Q_t \neq Q_c$*, and for fixed* B *and fixed* B_f *the rejection probabilities of SW-R, SW-F, ALL (always pool) and SW-PB all tend to zero. WU-R and the current-only test are consistent.*

The alternative in Theorem 3.15 is one for which the original condition $2(1-\gamma)D(Q_h, Q_c)\cos\beta < D(Q_c, Q_t)$ fails ($\cos\beta = 1$ and $D(Q_c, Q_t) = (1-\gamma)D(Q_c, Q_h)$). Full reselection repairs calibration, not power, and this example is not removed by recalibration.

Safeguard. To protect against such losses we combine the current-only and borrowed statistics over the same rows. For $r \in \{\text{cur}, \text{bor}\}$ let $p_{r,b} = L^{-1}\#\{a : S_a^r \geq S_b^r\}$ be the rank p-value of row b, $U_b = \min_r p_{r,b}$, and

$$p_G = \frac{1}{L}\#\{b : U_b \leq U_0\}. \tag{8}$$

This is the minP calibration of Westfall and Young (1993), and it differs from the unadjusted $\min(p_{\text{cur}}, p_{\text{bor}})$.

Proposition 3.16 (Safeguard)**.** *Under the conditions of Theorem 3.2,* $\mathbb{P}(p_G \leq \alpha) \leq \lfloor \alpha L \rfloor / L$*. For every data set* $p_G \leq 2\min(p_{\text{cur}}, p_{\text{bor}})$*; hence* $\mathbb{P}(p_G > \alpha) \leq \mathbb{P}(p_{\text{cur}} > \alpha/2)$ *and Theorem 3.12 with* $\delta = 0$ *and* $k_{\alpha/2} = \lfloor \alpha L/2 \rfloor \geq 1$ *bounds the Type-II error of the safeguarded test for every alternative and every borrowing rule.*

4 Synthetic Experiments

4.1 Design

All experiments use the Gaussian kernel with fixed bandwidth $\sigma = 1$ ($\kappa = 1$), $\alpha = 0.05$, $B = 199$ random permutations ($L = 200$, so $k_\alpha = 10$), $B_f = 199$ fusion bootstrap draws, and $B = 199$ partial-bootstrap draws for SW-PB. A fixed bandwidth keeps the radius θ meaningful across data sets and makes every rule label-invariant; the median heuristic is used only in the reproduction. Normal distributions are written $N(\text{mean}, \text{sd}^2)$, and the current control is always $Q_c = N(0, 1)$.

Reproduction of the original Table 2 (single source). $Q_t = N(0, 1)$, $Q_c = N(\mu_c - \mu_t, 1)$ with $\mu_c - \mu_t \in \{-0.4, 0, 0.4\}$, $Q_h = N(\mu_c + 0.2, 1)$, $m = 50$, $n = \ell = 100$, $\theta = 0.4$, $\alpha_f = 0.05$, $B = B_f = 999$, median-heuristic bandwidth on the pooled data (σ^2 = median squared distance), 1000 data sets per row. This repeats the original design as closely as the paper allows; Section 5.1 discusses the remaining differences.

Table 1: Guarantees of the procedures compared (fixed number B of permutations, $\alpha L \geq 1$). "Exact" means $\mathbb{P}_{H_0}(\text{reject}) \leq \lfloor \alpha L \rfloor / L$ for all sample sizes and all historical distributions.

Code	Procedure	Final level	Consistency vs. fixed $Q_t \neq Q_c$
CUR	current only	exact	all
ALL	pool all sources, partial permutation	exact	not all (Theorem 3.15)
WU-R	unbiased-risk weights, full reselection	exact (Theorem 3.2)	all, uniform $\sqrt{n}$ (Corollary 3.13)
WU-F	same weights, frozen	no guarantee	not established
WC-R	conservative-proxy weights, full reselection	exact	all (displacement $\leq \hat{v}_c/m$)
SW-R	sourcewise fusion, full reselection	exact	not all (Theorem 3.15)
SW-F / SW-PB	sourcewise fusion, frozen	$\leq \min(1, 2^J \alpha)$ (Proposition 3.3)	not all
SW-S, WU-S	sample splitting	exact (Proposition 3.11)	sufficient conditions only
G-WU, G-SW	minP with current only	exact	all (Proposition 3.16)

Main factorial experiment (A_main). $J = 3$ sources, $m = 40$, $n = 80$, $\ell_j = 80$, $\theta_j = 0.5$, Bonferroni fusion levels $\alpha_{f,j} = 0.05/3$. Source scenarios:

Scenario	$(Q_{h_1}, Q_{h_2}, Q_{h_3})$
Aligned	$N(0,1),\ N(0,1),\ N(0,1)$ (all identical to Q_c)
Partial	$N(0,1),\ N(0,1),\ N(1,1)$ (some identical)
Shifted	$N(0.5,1),\ N(0.7,1),\ N(1,1)$ (all different, same direction)
Opposite	$N(0.6,1),\ N(-0.6,1),\ N(1.5,1)$ (opposite directions: mean shifts cancel)
Mixed	$N(0,1),\ N(0,1.5^2),\ N(0.8,1)$ (identical, scale-shifted, location-shifted)

For reference, $D(N(0,1), N(\mu,1)) = \{2(1-e^{-\mu^2/6})/\sqrt{3}\}^{1/2}$ for $\sigma = 1$, which is $0.217, 0.300, 0.421$ for $\mu = 0.5, 0.7, 1$. Treatments: Null $Q_t = Q_c$; Mean+ $N(0.5,1)$ (same direction as the positive source shifts); Mean− $N(-0.5,1)$ (opposite direction); Scale $N(0,1.5^2)$; Shape $\frac{1}{2}N(-0.8,0.6^2) + \frac{1}{2}N(0.8,0.6^2)$ (same mean and variance as Q_c, bimodal). There are 2000 data sets per null setting and 1000 per alternative.

Sample-size experiment (A_n). $m \in \{20, 40, 80, 160\}$, $n = 2m$, $\ell_j = 2m$, $J = 3$, $\theta = 0.5$. Settings: AlignedNull (all sources $N(0,1)$, $Q_t = Q_c$); ShiftedNull (all sources $N(0.6,1)$, $D = 0.259 < \theta$); AlignedMean (aligned sources, $Q_t = N(0.4,1)$); and BlindSpot (all sources $N(0.6,1)$, $Q_t = \frac{1}{7}N(0,1) + \frac{6}{7}N(0.6,1)$). Here $\frac{1}{7} = m/(m + \ell_{\text{tot}})$, so pooling all three sources reproduces Q_t exactly (Theorem 3.15).

Procedures. Codes as in Table 1: CUR; ALL (sample-size pooling of all sources, partial permutation); WU-R and WU-F (unbiased-risk weights, grid step 0.05, $s \leq 0.9$, 1330 grid points, reselected or frozen); WC-R (conservative objective of the earlier draft, reselected); SW-R, SW-F, SW-PB (sourcewise fusion with reselection, frozen with partial permutation, frozen with partial bootstrap); SWu-R (SW-R without Bonferroni, $\alpha_{f,j} = 0.05$); SW-S and WU-S (sample splitting with half of the controls as pilot); G-WU and G-SW (minP safeguard).

Common random numbers. Within a data set all procedures share the permutations $A_1, \ldots, A_B$ and the bootstrap weight matrices. Differences between procedures are therefore paired, and we report paired 95% intervals for power differences.

4.2 Monte Carlo protocol

Seeds are derived with NumPy's `SeedSequence` from the tuple `(seed, setting, level, replicate)`, so each data set can be regenerated on its own. Exploratory runs use base seed 20261018. Confirmation runs repeat the complete designs with base seed 20261019 (2000/1000 replicates for A_main, 1000/500 for A_n). They were specified after the exploratory A_main results had been seen; because whole designs are repeated rather than selected cells, they check the stability of every reported pattern (Section 5.6). The exploratory results are reported unchanged. Rejection rates carry pointwise Wilson 95% intervals; in tables, a bold null rate with ↑ marks an interval lying entirely above 0.05. With 2000 replicates, a rate of 0.05 has Monte Carlo standard error 0.0049. Across 25 null cells and 13 procedures, a few marginal exceedances are expected by chance, so we interpret only systematic patterns. Code, raw arrays and environment are described in Appendix C.

5 Results

5.1 Reproduction of the original single-source experiment

Table 2: Reproduction of Yang et al. (2026, Table 2, median-heuristic row): merge rate and rejection rates, 1000 data sets per row ($m = 50$, $n = \ell = 100$, $\theta = 0.4$, $\mu_h - \mu_c = 0.2$). PB = partial bootstrap, PP = partial permutation; "frozen" is the original architecture, "reselect" repeats fusion in every permutation. Bold with ↑: Wilson 95% interval above 0.05.

$\mu_c - \mu_t$	merge	Current only	PB (orig.)	PP frozen (orig.)	PP reselect	Always pool	minP guard
-0.4	0.772	0.490	0.402	0.539	0.459	0.457	0.482
+0.0	0.776	0.048	**0.078**↑	**0.067**↑	0.046	0.050	0.049
+0.4	0.789	0.523	0.731	0.653	0.598	0.664	0.568

Original Table 2 (median heuristic): merge 0.285/0.307/0.272; PB 0.577/0.064/0.572; PP 0.513/0.051/0.535.

Table 2 shows that the qualitative findings are reproduced. Under the null ($\mu_c = \mu_t$), both frozen procedures of the original paper exceed the nominal level: 0.078 (PB) and 0.067 (PP), against 0.064 and 0.051 in the original. The full-reselection version of the same partial permutation gives 0.046. When $\mu_c - \mu_t = +0.4$ the historical shift points away from the treatment effect and borrowing raises power. When $\mu_c - \mu_t = -0.4$ the shift and the effect point the same way and borrowing lowers power. In that case the partial bootstrap has 0.402 against 0.490 for current-only testing, which is the cancellation behind Remark 3.19.

Our merge rate (0.77–0.79) is much higher than the original 0.27–0.31, so the absolute power values are not comparable. The paper states the bandwidth as the median heuristic but does not fix the kernel parameterisation unambiguously, and it uses a different bandwidth for the merged and unmerged branches. In Appendix B we repeat the experiment with the fixed bandwidths of the original Table 2 read as $k(x, y) = e^{-(x-y)^2/s^2}$. Under that reading our merge rates (0.24, 0.60, 0.81 for $s = 0.5, 1, 1.5$) move towards the original ones (0.34, 0.67, 0.82) (Table 6). The power under the alternatives, however, stays well below the original values (e.g. partial bootstrap at $\mu_c - \mu_t = -0.4$, $s = 1$: 0.407 against 0.572). The null rejection rates are again above 0.05 for the frozen procedures

Table 3: Null rejection rates (A_main, $m = 40$, 2000 data sets per row; Monte Carlo s.e. ≈ 0.005). Bold with ↑: Wilson 95% interval entirely above 0.05.

Sources	CUR	ALL	WU-R	WU-F	WC-R	SW-R	SW-F	SW-PB	SW-S	WU-S	G-WU	G-SW
Aligned	0.059	0.046	0.052	**0.095**↑	0.056	0.048	**0.061**↑	0.025	0.041	0.049	0.052	0.052
Partial	0.046	0.051	0.047	**0.114**↑	0.048	0.051	**0.063**↑	0.041	0.053	0.057	0.046	0.049
Shifted	0.045	0.041	0.045	**0.062**↑	0.049	0.049	**0.068**↑	**0.117**↑	0.046	0.049	0.045	0.048
Opposite	0.042	0.049	0.041	**0.126**↑	0.043	0.045	**0.085**↑	**0.139**↑	0.051	0.051	0.040	0.042
Mixed	0.056	0.057	0.057	**0.125**↑	0.056	0.054	**0.096**↑	**0.093**↑	0.043	0.048	0.053	0.054

at $s = 1.5$ (0.088, 0.064; original 0.074, 0.065). We could not identify the remaining difference, which may lie in the data-generating conventions or in the bandwidth used for the two branches. Our conclusions rest on the comparisons *within* our own runs, which share data and resampling.

5.2 Type-I error

Table 3 answers the calibration question directly.

- Every procedure whose level is guaranteed by Theorem 3.2 and Propositions 3.11 and 3.16 (CUR, ALL, WU-R, WC-R, SW-R, SW-S, WU-S, G-WU, G-SW) stays within Monte Carlo error of 0.05 in all five scenarios (0.040–0.059). This includes the Shifted and Opposite scenarios, where every source differs from Q_c.
- Freezing the selection inflates the size in every scenario. Frozen risk weights (WU-F) reach 0.062–0.126, because the weights take values on a grid of 1330 points and are fitted to the observed control sample; frozen sourcewise fusion (SW-F) reaches 0.061–0.096. The original partial bootstrap with a frozen selection (SW-PB) reaches 0.093–0.139 when merged sources are shifted (Shifted, Opposite, Mixed). It is conservative (0.025) when all merged sources are identical to Q_c, the degenerate case the original Theorem 3.5 excludes. Study B, Section 3.7, analyses this case.
- Bonferroni adjustment of the fusion stage does not repair the frozen tests: SW-F and SW-PB use Bonferroni levels throughout. Conversely, removing the adjustment (SWu-R, Appendix B) leaves the reselected test valid (0.047–0.054), as Proposition 3.8 predicts.

5.3 Power

Among level-α procedures, Table 4 and Fig. 1 show the following.

1. *Gains with compatible sources.* With three identical sources, WU-R improves on current-only testing by $+0.063 \pm 0.018$ (Mean+), $+0.077 \pm 0.020$ (Mean−) and $+0.110 \pm 0.024$ (Scale); SW-R gains +0.053, +0.061 and +0.037. ALL gains more (+0.13 to +0.26) because it uses the full sample-size weights, but ALL has no protection in any other scenario.
2. *Gains when the source shift opposes the effect.* In the Partial, Shifted and Mixed scenarios with Mean−, WU-R gains +0.097, +0.103 and +0.073. Shifted sources move the fused control away from the treatment and enlarge the contrast. This is a genuine but fragile gain, because it depends on the sign of an unknown effect.
3. *Losses when the source shift mimics the effect.* In Shifted with Mean+, WU-R loses -0.139 ± 0.021, SW-R loses -0.147 ± 0.023 and ALL collapses to 0.004. In Shifted and Opposite with

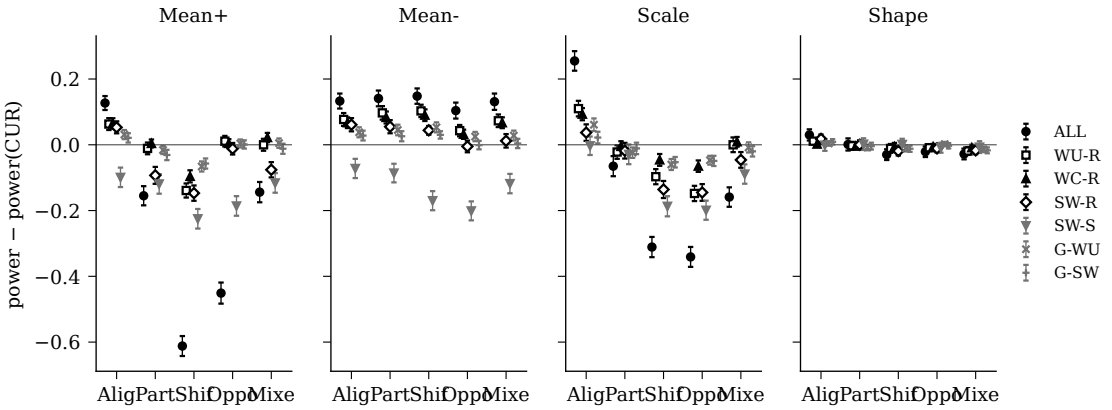


Figure 1: Paired power differences relative to current-only testing with 95% intervals (A_main). Scenario labels: Alig(ned), Part(ial), Shif(ted), Oppo(site), Mixe(d). Only procedures with guaranteed level are shown.

Scale, WU-R loses −0.097 and −0.148. The Opposite case is instructive. The Gram matrix sees that the ±0.6 mean shifts cancel and gives the two sources a combined weight of about 0.74 (Table 5). Their mixture has the right mean but variance 1.36, which moves the fused control towards the scale alternative. The risk objective is blind to which features of the distribution matter for the alternative at hand.

4. *The safeguard trades gains for protection.* G-WU halves both the gains (e.g. +0.031 instead of +0.063) and the losses (e.g. −0.066 instead of −0.139; −0.047 instead of −0.148). It was never more than 0.07 below current-only testing in this design.

5. *Sample splitting is costly.* SW-S and WU-S lose power in almost every cell. Halving the controls used for testing costs more than borrowing gains at $m = 40$.

6. *Shape alternative.* The bimodal alternative with the same mean and variance is barely detectable at this sample size with $\sigma = 1$ (rates 0.04–0.10 for procedures with guaranteed level). The frozen procedures appear more powerful, but this reflects their size inflation.

Why reselection costs power. Under an alternative, a permuted control sample is a mixture of Q_c and Q_t, so the fusion stage behaves differently in permuted rows than in the observed row. In Aligned/Mean+, identical sources are accepted in 95% of observed rows but only 71% of permuted rows. In Shifted/Mean+, the most shifted sources are accepted in 1% of observed rows but 16% of permuted rows (Table 7 in Appendix B). Under the null both rates coincide, which is exactly why reselection is exact. Under alternatives the reference distribution is built from rows that borrow differently from the observed row, and this dilutes the borrowed contrast. The frozen tests avoid this dilution and pay for it with size inflation.

5.4 Weights and selections

Table 5 shows that the unbiased objective borrows about 1.5 times more than the conservative objective of the earlier draft (total weight 0.83 versus 0.55 in Aligned), consistent with the double counting identified in Section 3.3. The extra borrowing translates into slightly higher power where borrowing helps (Aligned Scale 0.541 vs. 0.524) and slightly larger losses where it hurts (Opposite Scale 0.257 vs. 0.340). Weights respond to distances as intended. Sources at distance 0.42 or more

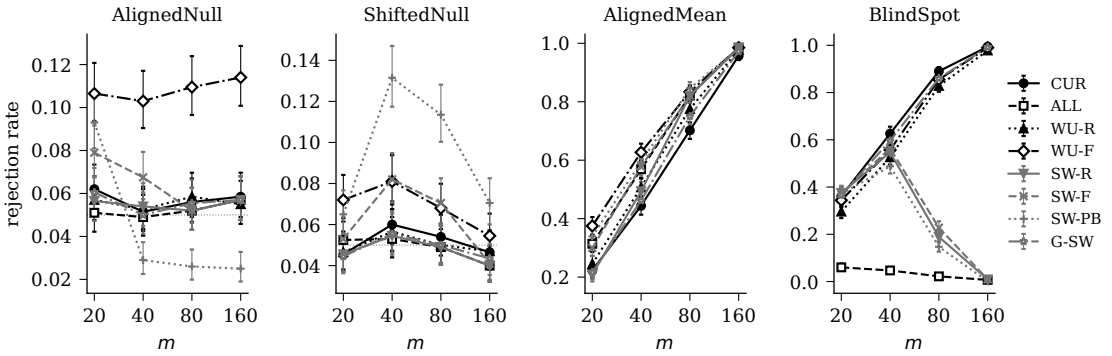


Figure 2: A_n: rejection rates against m ($n = \ell_j = 2m$, three sources, $\theta = 0.5$) with Wilson 95% intervals; dotted line at 0.05 in null panels. BlindSpot: all sources $N(0.6, 1)$ and $Q_t = \frac{1}{7}N(0, 1) + \frac{6}{7}N(0.6, 1)$, so pooling all sources reproduces Q_t. Full table in Table 8.

receive little weight (≤ 0.07), and the Partial scenario shifts weight onto the two identical sources. The sourcewise rule accepts each identical source in about 95% of data sets and the $N(0.5, 1)$ source in 43%. The deterministic bound of Theorem 3.6 held in all 30,000 data sets and all 200 rows of each, as the code checks.

5.5 Growing samples and the blind spot

Figure 2 and Table 8 show three patterns.

Frozen selection: which inflation vanishes. SW-F rejected at 0.079 at $m = 20$ and 0.057 at $m = 160$ under AlignedNull, and at 0.082 ($m = 40$) and 0.041 ($m = 160$) under ShiftedNull. This matches the oracle-coupling mechanism: with a fixed radius and fixed distributions, the sourcewise selection stabilises (identical and $N(0.6, 1)$ sources are eventually always accepted), and the conditioning effect fades. Frozen *risk weights* (WU-F) stayed at 0.10–0.11 under AlignedNull for every m. Soft weights never stabilise: $m\hat{D}^2/\hat{v}_c$ has a nondegenerate limit, so the selected weights remain random at every sample size. The original partial bootstrap (SW-PB) was anti-conservative at small m (0.093, 0.132) and, when merged sources are identical, conservative at larger m (0.025–0.029). With shifted merged sources it stayed above 0.05 up to $m = 160$ (0.070).

Reselected and fixed-rule procedures stay calibrated. Across the 8 null cells, CUR, ALL, WU-R and SW-R stayed between 0.040 and 0.062. Two cells marked ↑ for CUR (an exact test) and one for G-SW are within the range expected from about 40 marginal intervals. As a separate check, 20,000 additional current-only tests with $m = 20$ gave 0.0506 (s.e. 0.0015).

The blind spot is real and is removed by risk weighting. In BlindSpot, ALL has power 0.060, 0.047, 0.022, 0.007 as m grows, and SW-R, SW-F and SW-PB fall to 0.009, 0.009 and 0.001 at $m = 160$, while current-only testing reaches 0.994. This is Theorem 3.15 in action: for $m \geq 80$ every source passes the fixed-radius fusion test, and the pooled control equals Q_t. WU-R follows the current-only curve from slightly below (0.296 vs. 0.361 at $m = 20$; 0.978 vs. 0.994 at $m = 160$), as Corollary 3.13 predicts, and G-SW repairs SW-R (0.992). With aligned sources, SW-R gains up to 0.12 ($m = 80$: 0.820 vs. 0.702) and WU-R up to 0.07.

Cost. Per data set, SW-R took 0.10, 0.16, 0.20 and 0.89 s at $m = 20, 40, 80, 160$ (row-wise fusion bootstrap), and WU-R took 0.09–0.17 s.

5.6 Independent-seed confirmation

Confirmation runs pending.

5.7 Computation

Per data set at $m = 40$ on one worker: shared Gram matrix and allocation statistics 0.016 s; CUR and ALL under 1 ms; WU-R 0.18 s (1330 grid points × 200 rows); SW-R 0.17 s (bootstrap quadratic forms for 200 rows × 199 draws); SW-PB 0.03 s; the safeguard adds 3 ms. The cost of full reselection is dominated by the per-row fusion bootstrap, which is $O(LB_f m^2)$; at $m = 160$ it was 0.89 s per data set (Section 5.5). A label-invariant pooled bootstrap, as used in Study B, removes the factor L.

5.8 Discussion and conclusions

What is established. (i) Selection must be part of the calibration. Freezing an observed selection led to Type-I errors up to 0.14 in our designs, and for soft weights the excess did not shrink with m, while re-running the entire selection in every permutation of the current labels gives exact finite-sample level for any symmetric rule (Theorem 3.2), and this was confirmed empirically in all settings. (ii) The unbiased risk objective, with zero borrowing among the candidates, moves the fused control by at most $O(m^{-1/2})$ in every row, deterministically. This yields uniform $\sqrt{n}$-rate consistency over all historical distributions (Corollary 3.13) and removes the fixed-radius blind spot. (iii) Sourcewise fusion with a fixed radius keeps a blind spot even with exact calibration (Theorem 3.15).

What is not established. We have no theorem saying that borrowing increases power, and the experiments show it does not always: losses of up to 0.15 occurred when the historical shift resembled the treatment effect, in location or in scale. Consistency, a property as $n \to \infty$, should not be read as an improvement at the sample size of a trial. Embedding-risk reduction is not power. The risk objective measures closeness to Q_c in the RKHS norm and does not know which direction the alternative takes.

Practical reading. If borrowing must be justified to a regulator, use full reselection or sample splitting, never a frozen selection. Prefer the risk weights with the minP safeguard when the direction of possible historical drift relative to the treatment effect is unknown. Expect gains of roughly 3–8 percentage points with compatible sources and limited losses otherwise, at the sample sizes we studied. Without the safeguard, gains and losses both roughly double.

Open problems. Local asymptotic power of reselected tests; borrowing rules whose objective targets the test statistic rather than the embedding risk; increasing J; joint calibration of radius and weights; covariate-adjusted versions; and real-data validation. Our novelty assessment is limited to the literature listed in Section 2.4; individual ingredients (random-permutation exactness, minP calibration, Stein-type shrinkage) are established, and our contribution is their combination and analysis for this problem.

A Proofs

A.1 Proof of Theorem 3.2 and of the level part of Proposition 3.16

Lemma A.1 (Rank lemma)**.** *Let $S_0, \dots, S_B$ be exchangeable, $L = B + 1$, and $R_b = \#\{a : S_a \geq S_b\}$. For every integer $k \geq 0$, $\mathbb{P}(R_0 \leq k) \leq k/L$.*

Proof. Let $I = \{b : R_b \leq k\}$. If $I \neq \emptyset$, choose $b^* \in I$ with the smallest value S_{b^*} among I. Every $b \in I$ satisfies $S_b \geq S_{b^*}$, so $I \subseteq \{a : S_a \geq S_{b^*}\}$, a set of size $R_{b^*} \leq k$. Hence $\sum_b \mathbf{1}\{R_b \leq k\} \leq k$ deterministically. By exchangeability $\mathbb{P}(R_b \leq k)$ does not depend on b, so $\mathbb{P}(R_0 \leq k) = L^{-1}\mathbb{E}\sum_b \mathbf{1}\{R_b \leq k\} \leq k/L$. □

Proof of Theorem 3.2. Write $Z = \sigma(O)$, where O is the multiset of pooled values and σ the arrangement of those values in positions. Under H_0 the coordinates of Z are i.i.d., so conditionally on O (and on $\mathbb{H}, U$, which are independent of Z) the arrangement is uniform. Let V_b be the multiset of values at positions A_b. Then $V_0 = \sigma(A_0)$ is a uniform m-sub-multiset of O. For $b \geq 1$, conditionally on σ, $V_b = \sigma(A_b)$ are i.i.d. uniform because A_b are i.i.d. uniform and independent of σ; hence $V_1, \dots, V_B$ are i.i.d. uniform and independent of σ, in particular of V_0. So $V_0, \dots, V_B$ are i.i.d. given $(O, \mathbb{H}, U)$. By Assumption 3.1(ii) the row statistic is $S_b = h(V_b; O, \mathbb{H}, U)$ for one measurable h: the treatment multiset is $O \setminus V_b$, and the weights depend on the control sample only through V_b. Therefore $S_0, \dots, S_B$ are conditionally i.i.d., hence exchangeable. Now $p_{\mathrm{FR}} \leq \alpha$ iff $R_0 \leq \lfloor \alpha L \rfloor$, and Lemma A.1 gives the conditional bound $\lfloor \alpha L \rfloor / L$; integrating over $(O, \mathbb{H}, U)$ completes the proof. If the auxiliary randomness is drawn independently in each row (U_b i.i.d.), the same argument applies to the i.i.d. pairs (V_b, U_b). □

Level part of Proposition 3.16. The vector $(U_0, \dots, U_B)$ is obtained from $(S_b^{\mathrm{cur}}, S_b^{\mathrm{bor}})_b$ by a map that commutes with permutations of the row index, so it is exchangeable when the pairs are. Apply Lemma A.1 to $-U_b$: $p_G \leq \alpha$ iff $\#\{b : U_b \leq U_0\} \leq \lfloor \alpha L \rfloor$. □

A.2 Proofs of Propositions 3.3, 3.4 and 3.11

Proof of Proposition 3.3. $\{p_{\hat{d}} \leq \alpha\} = \bigcup_d \{p_d \leq \alpha, \hat{d} = d\}$, so $\mathbb{P}(p_{\hat{d}} \leq \alpha) \leq \sum_d \mathbb{P}(p_d \leq \alpha)$, and each fixed-rule p_d satisfies $\mathbb{P}(p_d \leq \alpha) \leq \alpha$ by Theorem 3.2 with a constant rule. □

Proof of Proposition 3.4. Let $\gamma = m/(m + \ell)$. The observed statistic $D^2(\gamma Q_c^m + (1 - \gamma)Q_h^\ell, Q_t^n)$ converges in probability to $(1 - \gamma)^2 D^2(Q_h, Q_c) > 0$ because each empirical embedding converges in $\mathcal{H}_k$ (bounded kernel, Hilbert-space law of large numbers). For a uniformly random allocation of all $M = m + \ell + n$ values into groups of sizes $m + \ell$ and n, the finite-population identity (9) below, with (m, n, N) replaced by $(m+\ell, n, M)$, gives $\mathbb{E}[D^2 \mid \text{data}] \leq \kappa^2 M^2/\{(m+\ell)nM - (m+\ell)n\} = O(n^{-1})$. Each of the B permuted statistics therefore tends to 0 in probability, so with probability tending to one all of them are below the observed statistic, giving $p = 1/L \leq \alpha$. □

Proof of Proposition 3.11. Conditionally on the pilot controls and $\mathbb{H}$, the selected weights are fixed, and the test controls and treated units are i.i.d. from $Q_c = Q_t$ and independent of the pilot. Theorem 3.2 with a constant rule applies conditionally, and integrating gives the claim. □

A.3 Proofs for the risk-weighted rule

Proof of Proposition 3.5. Write $e_c = \hat{\xi}_c - \xi_c$ and $e_j = \hat{\xi}_j - \xi_j$. These are independent, mean zero (Bochner expectation), with $\mathbb{E}\|e_c\|^2 = v_c/m$ and $\mathbb{E}\|e_j\|^2 = v_j/\ell_j$, since the off-diagonal terms of $\|m^{-1}\sum_i\{k(\cdot,x_i)-\xi_c\}\|^2$ have mean zero. Then

$$\hat{\xi}_f(w) - \xi_c = \sum_j w_j b_j + (1-s)e_c + \sum_j w_j e_j,$$

and expanding the square, all cross terms have mean zero, which gives $R(w)$. Similarly $\hat{\xi}_i - \hat{\xi}_c = b_i + e_i - e_c$ gives $\mathbb{E}\hat{G}_{ij} = \langle b_i, b_j\rangle + \mathbb{E}\langle e_i, e_j\rangle + \mathbb{E}\|e_c\|^2 = \langle b_i, b_j\rangle + \mathbf{1}\{i=j\}v_i/\ell_i + v_c/m$. For $\hat{v}_c$, $\mathbb{E}\|\hat{\xi}_c\|^2 = \|\xi_c\|^2 + v_c/m$, so $\mathbb{E}\hat{v}_c = \frac{m}{m-1}(v_c - v_c/m) = v_c$. Finally $\mathbb{E}w^\top\hat{G}w = \|\sum_j w_j b_j\|^2 + s^2 v_c/m + \sum_j w_j^2 v_j/\ell_j$, and $R(w) - \mathbb{E}w^\top\hat{G}w = \{(1-s)^2 - s^2\}v_c/m = (1-2s)v_c/m$. □

Proof of Theorem 3.6. $\hat{R}_U(0) = \hat{v}_c/m$, so $\hat{R}_U(\hat{w}) \le \hat{R}_U(0)$ reads $\hat{w}^\top\hat{G}\hat{w} + (1-2\hat{s})\hat{v}_c/m \le \hat{v}_c/m$, i.e. $\hat{w}^\top\hat{G}\hat{w} \le 2\hat{s}\hat{v}_c/m$. Since $\hat{v}_c = \frac{m}{m-1}\{m^{-1}\sum_i k(x_i,x_i) - \|\hat{\xi}_c\|^2\} \le \frac{m}{m-1}\kappa^2$, we get $\hat{v}_c/m \le \kappa^2/(m-1)$. The identity $\hat{w}^\top\hat{G}\hat{w} = \|\hat{\xi}_f(\hat{w}) - \hat{\xi}_c\|^2$ is (2). Nothing in the argument uses the distribution of the data, so it applies to every row. □

Proof of Proposition 3.7. By Minkowski's inequality in $L^2(\mathbb{P};\mathcal{H}_k)$,

$$\left(\mathbb{E}\|\hat{\xi}_f(\hat{w}) - \xi_c\|^2\right)^{1/2} \le \left(\mathbb{E}\|\hat{\xi}_f(\hat{w}) - \hat{\xi}_c\|^2\right)^{1/2} + \left(\mathbb{E}\|\hat{\xi}_c - \xi_c\|^2\right)^{1/2};$$

apply Theorem 3.6 to the first term and $\mathbb{E}\|e_c\|^2 = v_c/m \le \kappa^2/(m-1)$ to the second. □

A.4 Proofs for sourcewise fusion

Proof of Proposition 3.8. The first inequality is a union bound over $j \in \mathcal{J}_0$ and needs no independence between the tests, which share the control sample. The limit statement follows termwise because J is fixed. □

Proof of Lemma 3.9. Every accepted j satisfies $D(Q^m_{c,A_b}, Q^{\ell_j}_{h_j}) < \theta_j - q_j \le \theta_j$ because $q_j \ge 0$. With sample-size weights, $\hat{\xi}_{f,b} - \hat{\xi}_{c,b} = \sum_{j\in\hat{S}_b} w_j(\hat{\xi}_j - \hat{\xi}_{c,b})$, so by the triangle inequality its norm is at most $\sum_{j\in\hat{S}_b} w_j\theta_j \le \max_j\theta_j \cdot \ell_{\hat{S}_b}/(m+\ell_{\hat{S}_b}) \le \rho\max_j\theta_j$. □

Proof of Lemma 3.10. For a sample $a_1,\dots,a_M$ with embedding $\bar{a}$ and Efron weights W,

$$\mathbb{E}^*\left\|\frac{1}{M}\sum_i (W_i - 1)k(\cdot,a_i)\right\|^2 = \frac{1}{M}\left\{\frac{1}{M}\sum_i k(a_i,a_i) - \|\bar{a}\|^2\right\} \le \frac{\kappa^2}{M}.$$

Hence $\mathbb{E}(S^{(r)})^2 \le 2\kappa^2(1/m + 1/\ell_j)$. At least $B_f - k_f + 1$ of the $S^{(r)}$ are $\ge q_j$, so $(B_f - k_f + 1)q_j^2 \le \sum_r (S^{(r)})^2$; take expectations. The same argument with M = pool size covers the label-invariant pooled variant used in Study B. □

A.5 Proofs for power

Lemma A.2 (Finite-population identity)**.** *For fixed values $Z_1,\dots,Z_N$ and a uniform random m-subset A, with $a_i = k(\cdot,Z_i) - N^{-1}\sum_{i'} k(\cdot,Z_{i'})$,*

$$\mathbb{E}_A\|\hat{\xi}_{c,A} - \hat{\xi}_{t,A}\|^2 = \frac{N}{mn(N-1)}\sum_{i=1}^N \|a_i\|^2 \le M_{m,n}. \tag{9}$$

Proof. Since $\sum_i a_i = 0$, $\hat{\xi}_{c,A} - \hat{\xi}_{t,A} = (m^{-1}+n^{-1})\sum_{i\in A} a_i = \frac{N}{mn}\sum_i I_i a_i$ with $I_i = \mathbf{1}\{i \in A\}$. Using $\mathbb{E}I_i = m/N$, $\mathbb{E}I_i I_{i'} = m(m-1)/\{N(N-1)\}$ for $i \neq i'$, and $\sum_{i\neq i'}\langle a_i, a_{i'}\rangle = -\sum_i \|a_i\|^2$,

$$\mathbb{E}\Big\|\sum_i I_i a_i\Big\|^2 = \Big\{\frac{m}{N} - \frac{m(m-1)}{N(N-1)}\Big\}\sum_i \|a_i\|^2 = \frac{mn}{N(N-1)}\sum_i \|a_i\|^2.$$

Multiply by $N^2/(mn)^2$. Finally $\sum_i \|a_i\|^2 = \sum_i k(Z_i, Z_i) - N\|\bar{\xi}\|^2 \le N\kappa^2$. □

Proof of Theorem 3.12. Write $\hat{\Delta} = \|\hat{\xi}_c - \hat{\xi}_t\|$ (observed allocation) and $R_b = \|\hat{\xi}_{c,A_b} - \hat{\xi}_{t,A_b}\|$. In every row the triangle inequality and the displacement condition give

$$\hat{\Delta} - \delta \le \sqrt{S_0}, \qquad \sqrt{S_b} \le R_b + \delta.$$

The test does not reject iff $K := \#\{b \ge 1 : S_b \ge S_0\} \ge k_\alpha$. Let $\varepsilon = \tau = (d - 2\delta)/2 > 0$ and $E = \{\hat{\Delta} \ge d - \varepsilon\}$. On E, $S_b \ge S_0$ implies $R_b \ge d - \varepsilon - 2\delta = \tau$, so $K \le K' := \#\{b \ge 1 : R_b \ge \tau\}$. Hence

$$\mathbb{P}(p_{\mathrm{FR}} > \alpha) \le \mathbb{P}(E^c) + \mathbb{P}(K' \ge k_\alpha) \le \frac{\mathbb{E}\|(\hat{\xi}_t - \hat{\xi}_c) - (\xi_t - \xi_c)\|^2}{\varepsilon^2} + \frac{\mathbb{E}K'}{k_\alpha}.$$

The first numerator equals $v_t/n + v_c/m \le v_{m,n}$. Conditionally on the data the A_b are uniform, so by Markov's inequality and Lemma A.2, $\mathbb{E}[K' \mid Z] \le B\, M_{m,n}/\tau^2$. Substituting $\varepsilon = \tau = (d - 2\delta)/2$ gives the bound. □

Proof of Corollaries 3.13 and 3.14. For Corollary 3.13, Theorem 3.6 holds in every row, so Theorem 3.12 applies with $\delta_m = \kappa\sqrt{2/(m-1)}$, and the bound depends on the distributions only through d. Since $\sqrt{n}\delta_m = O(1)$ and $\sqrt{n}d_n \to \infty$, we have $n(d_n - 2\delta_m)^2 \to \infty$ and the bound tends to zero uniformly. For Corollary 3.14 use Lemma 3.9. □

Proof of Theorem 3.15. $Q_t \neq Q_c$ because $\xi_t - \xi_c = (1-\gamma)(\xi_h - \xi_c) \neq 0$. Write $D_h = D(Q_c, Q_h)$, $\lambda = m/N$ and $\xi_d = \lambda\xi_c + (1-\lambda)\xi_t$.

Row 0. $D(Q_c^m, Q_{h_j}^{\ell_j}) \to D_h < \theta$ and $q_j \to 0$ in probability (Lemma 3.10), so every source is accepted with probability tending to one. Then $\hat{\xi}_f = \gamma\hat{\xi}_c + (1-\gamma)\ell_{\mathrm{tot}}^{-1}\sum_j \ell_j\hat{\xi}_j \to \gamma\xi_c + (1-\gamma)\xi_h = \xi_t$, so $S_0 \to 0$.

Rows $b \ge 1$. By Lemma A.2 applied to the pool and its complement, $\hat{\xi}_{c,A_b}$ and $\hat{\xi}_{t,A_b}$ both converge in probability to the pooled embedding, which converges to ξ_d. Hence $D(Q_{c,A_b}^m, Q_{h_j}^{\ell_j}) \to \|\xi_d - \xi_h\|$, where

$$\xi_d - \xi_h = \{\lambda + (1-\lambda)\gamma\}(\xi_c - \xi_h),$$

so $\|\xi_d - \xi_h\| = \{\lambda + (1-\lambda)\gamma\}D_h < \theta$. Every source is accepted in every row with probability tending to one, and $S_b \to \|\gamma\xi_d + (1-\gamma)\xi_h - \xi_d\|^2 = (1-\gamma)^2\{\lambda + (1-\lambda)\gamma\}^2 D_h^2 =: c_* > 0$.

With B fixed, $\mathbb{P}(S_b > S_0\ \forall b \ge 1) \to 1$, on which event $p_{\mathrm{FR}} = 1 > \alpha$. The same holds for SW-F (row-0 selection) and for ALL. For SW-PB, $\Delta = \sqrt{n}\{S_0 - \|\hat{\xi}_f - \hat{\xi}_c\|^2\}$ with $\|\hat{\xi}_f - \hat{\xi}_c\|^2 \to (1-\gamma)^2 D_h^2 > 0$, so $\Delta \to -\infty$. The bootstrap statistic is $\sqrt{n}\{-2\langle A^*, B^*\rangle + \|B^*\|^2\}$ with $\|A^*\| \le 2\kappa$ and $B^* = O_P(n^{-1/2})$, so the critical value is $O_P(1)$. If $\hat{S} = \emptyset$ the current-only branch is used, which has probability tending to zero. WU-R is consistent by Corollary 3.13, and the current-only test by Theorem 3.12 with $\delta = 0$. □

Proof of Proposition 3.16, power part. For any vector T and $u \in [0,1]$, Lemma A.1's counting step gives $\#\{b : p_{r,b} \le u\} \le \lfloor uL \rfloor$. With $u = U_0 = \min_r p_{r,0}$, $\#\{b : U_b \le u\} \le \sum_r \#\{b : p_{r,b} \le u\} \le 2uL$, so $p_G \le 2U_0$. If $p_G > \alpha$ then $p_{\text{cur}} > \alpha/2$. The current-only test at level $\alpha/2$ rejects iff $\#\{b \ge 1 : S_b^{\text{cur}} \ge S_0^{\text{cur}}\} \le k_{\alpha/2} - 1$, and the proof of Theorem 3.12 with $\delta = 0$ and $k_{\alpha/2}$ in place of k_α applies. □

B Additional results

B.1 Bandwidth convention in the reproduction

B.2 Fusion levels and selection in permuted rows

Under every null, acceptance rates in observed and permuted rows agree to two decimals, as exchangeability requires. Under alternatives they differ, sometimes strongly. Dropping the Bonferroni adjustment keeps the level and changes power by a few points in either direction. It raises power when the extra accepted sources are compatible (Aligned) and lowers it when they are shifted towards the treatment (Shifted, Mean+: 0.391 versus 0.469).

B.3 Sample-size experiment: full table

C Implementation and reproducibility

Code. All code is in the repository `https://github.com/Yu-Jusang/ttp-extensions` (directory `code/`) and in the accompanying archive.

- `ttp.py`: kernel, Gram-matrix bookkeeping, allocations, row statistics, p-values (ties counted, observed row included), minP calibration, risk-weight selection, equivalence-fusion bootstrap (row-wise with value-sorted common weights, or label-invariant pooled), partial bootstrap.
- `methods.py`: the procedures of Study A (`study_A`) and Study B (`study_B`), sharing data, permutations and bootstrap weights within a data set.
- `run_sims.py`: suites `repro`, `repro_bw`, `A_main`, `A_n`, `B_n`, `B_local`; `run_all.sh` (exploratory, base seed 20261018) and `run_conf.sh` (confirmation, base seed 20261019).
- `analyze.py`: all tables (LaTeX fragments in `paper_*/gen/`) and figures from the raw arrays.
- `pb_scalar.py`: the Gaussian-limit calculation for the partial bootstrap at degeneracy (Study B).

How to run. From the repository root:

```
cd code
bash run_all.sh
bash run_conf.sh
python analyze.py
```

Each suite writes `results/<suite>.npz`, which holds one array of p-values (or 0/1 decisions for partial-bootstrap procedures), timings and diagnostics per setting and procedure. It also writes a metadata file `results/<suite>_meta.json` recording the seed, replicate counts, elapsed time and software versions. Every data set is regenerated by NumPy's `SeedSequence` from `(seed, setting, level, replicate)`.

Environment. Windows 11, Intel Core i7-1360P (12 cores, 16 threads; 15 worker processes, one BLAS thread each), Python 3.12.2, NumPy 2.5.4, SciPy 1.18.1, Matplotlib 3.11.2, TeX Live 2024. Reported times are per data set on one worker and include only the step named.

Implementation details. (i) All statistics are quadratic forms in one Gram matrix K of the $N+\sum_j \ell_j$ outcomes. A permutation row is a pair of weight vectors $(\mathbf{1}_A/m, \mathbf{1}_{A^c}/n)$, and all L rows are processed by one matrix product. (ii) Rows $b \geq 1$ are uniform m-subsets drawn with replacement across rows; the observed allocation is row 0. (iii) Ties: $p = L^{-1}\#\{b : S_b \geq S_0 - 10^{-12}\max(1, |S_0|)\}$. (iv) The fusion quantile is the $\lceil(1-\alpha_f)B_f\rceil$-th order statistic. (v) The multinomial bootstrap weights of the current control are attached to the controls sorted by value (the symmetry assumption, Assumption 3.1 of Study A). (vi) The risk-weight grid contains 0 and is sorted by total weight, so ties are broken towards less borrowing.

What was not done. We did not analyse real data. The Prospera application of the original paper was not repeated because its data are not distributed with the paper, and the original code was not available to us. Differences from the original Table 2 are discussed in the reproduction section.

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Table 4: Rejection rates under alternatives (A_main, $m = 40$, $n = 80$, $\ell_j = 80$; 1000 data sets per row). Bold: at least 0.03 above CUR; italic: at least 0.03 below CUR. Frozen procedures (WU-F, SW-F, SW-PB) do not control the level (Table 3), so their power is not comparable.

Treatment	Sources	CUR	ALL	WU-R	WU-F	WC-R	SW-R	SW-F	SW-PB	SW-S	WU-S	G-WU	G-SW
Mean+	Aligned	0.602	**0.729**	**0.665**	**0.755**	**0.667**	**0.655**	**0.731**	**0.755**	*0.503*	*0.543*	**0.633**	0.623
	Partial	0.611	*0.456*	0.600	**0.767**	0.615	*0.518*	**0.749**	**0.753**	*0.492*	*0.507*	0.595	*0.580*
	Shifted	0.616	*0.004*	*0.477*	*0.545*	*0.520*	*0.469*	*0.574*	*0.488*	*0.391*	*0.299*	*0.550*	*0.559*
	Opposite	0.592	*0.141*	0.603	**0.722**	0.596	0.580	0.587	0.570	*0.406*	*0.483*	0.596	0.593
	Mixed	0.573	*0.429*	0.573	**0.731**	0.595	*0.497*	**0.718**	**0.711**	*0.457*	*0.496*	0.578	0.561
Mean-	Aligned	0.600	**0.733**	**0.677**	**0.758**	**0.667**	**0.661**	**0.738**	**0.774**	*0.528*	*0.557*	**0.638**	0.628
	Partial	0.599	**0.740**	**0.696**	**0.758**	**0.682**	**0.654**	**0.720**	**0.774**	*0.513*	*0.566*	**0.644**	0.625
	Shifted	0.592	**0.740**	**0.695**	**0.729**	**0.682**	**0.636**	**0.642**	**0.654**	*0.422*	*0.548*	**0.645**	**0.622**
	Opposite	0.618	**0.722**	**0.661**	**0.748**	**0.650**	0.613	0.607	*0.574*	*0.417*	*0.527*	0.643	0.617
	Mixed	0.587	**0.718**	**0.660**	**0.763**	**0.654**	0.599	**0.730**	**0.746**	*0.469*	*0.525*	0.615	0.591
Scale	Aligned	0.431	**0.686**	**0.541**	**0.767**	**0.524**	**0.468**	**0.705**	**0.675**	0.428	**0.487**	**0.492**	0.453
	Partial	0.440	*0.375*	0.418	**0.772**	0.435	0.422	**0.693**	**0.669**	0.411	0.454	0.416	0.429
	Shifted	0.426	*0.115*	*0.329*	**0.487**	*0.380*	*0.290*	**0.461**	**0.483**	*0.239*	*0.247*	*0.366*	*0.373*
	Opposite	0.405	*0.064*	*0.257*	**0.649**	*0.340*	*0.260*	**0.460**	**0.482**	*0.206*	*0.284*	*0.358*	*0.356*
	Mixed	0.438	*0.279*	0.438	**0.725**	0.447	*0.392*	**0.633**	**0.531**	*0.349*	*0.392*	0.430	0.418
Shape	Aligned	0.066	**0.096**	0.077	**0.161**	0.068	0.084	**0.112**	0.076	0.065	0.078	0.070	0.075
	Partial	0.063	0.064	0.060	**0.171**	0.059	0.063	**0.105**	0.082	0.065	0.074	0.055	0.059
	Shifted	0.070	*0.040*	0.059	0.085	0.064	0.051	0.086	**0.121**	0.071	0.057	0.059	0.058
	Opposite	0.059	0.038	0.050	**0.117**	0.053	0.048	0.085	**0.148**	0.050	0.046	0.061	0.056
	Mixed	0.064	0.036	0.045	**0.147**	0.055	0.047	**0.105**	**0.100**	0.057	0.064	0.050	0.047

Table 5: Mean selected weights (observed row) for the unbiased (WU) and conservative (WC) objectives, and sourcewise acceptance rates (SW, Bonferroni $\alpha_{f,j} = 0.05/3$).

		mean $\hat{w}_j$ (WU)			mean $\hat{w}_j$ (WC)			accept rate (SW)		
Sources	Treatment	h_1	h_2	h_3	h_1	h_2	h_3	h_1	h_2	h_3
Aligned	Null	0.28	0.29	0.26	0.18	0.19	0.18	0.94	0.95	0.96
Aligned	Mean+	0.27	0.29	0.28	0.18	0.18	0.18	0.95	0.96	0.95
Partial	Null	0.39	0.38	0.07	0.25	0.25	0.05	0.95	0.95	0.01
Partial	Mean+	0.40	0.38	0.06	0.26	0.25	0.05	0.95	0.96	0.00
Shifted	Null	0.25	0.03	0.00	0.16	0.02	0.00	0.43	0.13	0.01
Shifted	Mean+	0.24	0.03	0.00	0.16	0.02	0.00	0.40	0.12	0.01
Opposite	Null	0.36	0.38	0.01	0.23	0.24	0.01	0.25	0.25	0.00
Opposite	Mean+	0.35	0.37	0.01	0.22	0.24	0.01	0.23	0.24	0.00
Mixed	Null	0.62	0.10	0.08	0.37	0.08	0.06	0.96	0.43	0.06
Mixed	Mean+	0.62	0.10	0.08	0.37	0.08	0.06	0.96	0.42	0.05

Table 6: Original Table 2 rows with fixed bandwidth s, read as $k(x, y) = e^{-(x-y)^2/s^2}$ (i.e. $\sigma = s/\sqrt{2}$ in our parameterisation); 1000 data sets per cell, frozen selection as in the original. "original" rows copy Yang et al. (2026, Table 2): merge rates averaged over the three columns; PB with PP in the next column.

s	merge	$\mu_c - \mu_t = -0.4$		$\mu_c = \mu_t$		$\mu_c - \mu_t = +0.4$	
		PB	PP	PB	PP	PB	PP
0.5	0.235	0.338	0.358	0.055	0.059	0.398	0.386
original	0.34	0.528	0.511	0.070	0.063	0.536	0.566
1.0	0.598	0.407	0.504	0.062	0.058	0.626	0.569
original	0.67	0.572	0.559	0.074	0.053	0.711	0.629
1.5	0.806	0.398	0.584	**0.088**$^{\uparrow}$	**0.064**$^{\uparrow}$	0.694	0.625
original	0.82	0.581	0.566	0.074	0.065	0.739	0.677

Table 7: Sourcewise fusion with Bonferroni levels (SW-R, $\alpha_{f,j} = 0.05/3$) and without (SWu-R, $\alpha_{f,j} = 0.05$), both with full reselection; acceptance rates of each source in the observed row and averaged over permuted rows (A_main).

Sources	Treatment	SW-R	SWu-R	accept (obs. row) h_1, h_2, h_3	accept (perm. rows) h_1, h_2, h_3
Aligned	Null	0.048	0.048	0.94 0.95 0.96	0.95 0.96 0.96
Aligned	Mean+	0.655	0.689	0.95 0.96 0.95	0.70 0.71 0.71
Aligned	Mean-	0.661	0.690	0.96 0.95 0.96	0.72 0.71 0.71
Aligned	Scale	0.468	0.560	0.95 0.95 0.96	0.70 0.67 0.70
Aligned	Shape	0.084	0.092	0.96 0.95 0.95	0.93 0.93 0.93
Partial	Null	0.051	0.047	0.95 0.95 0.01	0.95 0.95 0.01
Partial	Mean+	0.518	0.563	0.95 0.96 0.00	0.72 0.72 0.17
Partial	Mean-	0.654	0.679	0.96 0.96 0.01	0.68 0.69 0.00
Partial	Scale	0.422	0.490	0.96 0.95 0.00	0.69 0.69 0.01
Partial	Shape	0.063	0.074	0.96 0.96 0.00	0.93 0.93 0.01
Shifted	Null	0.049	0.048	0.43 0.13 0.01	0.42 0.13 0.01
Shifted	Mean+	0.469	0.391	0.40 0.12 0.01	0.89 0.64 0.16
Shifted	Mean-	0.636	0.666	0.42 0.12 0.00	0.04 0.00 0.00
Shifted	Scale	0.290	0.234	0.43 0.13 0.01	0.31 0.13 0.01
Shifted	Shape	0.051	0.052	0.40 0.10 0.00	0.41 0.13 0.01
Opposite	Null	0.045	0.048	0.25 0.25 0.00	0.26 0.26 0.00
Opposite	Mean+	0.580	0.561	0.23 0.24 0.00	0.78 0.01 0.00
Opposite	Mean-	0.613	0.599	0.27 0.25 0.00	0.02 0.80 0.00
Opposite	Scale	0.260	0.204	0.27 0.26 0.00	0.20 0.21 0.00
Opposite	Shape	0.048	0.052	0.26 0.24 0.00	0.26 0.26 0.00
Mixed	Null	0.054	0.054	0.96 0.43 0.06	0.95 0.43 0.05
Mixed	Mean+	0.497	0.534	0.96 0.42 0.05	0.70 0.33 0.49
Mixed	Mean-	0.599	0.637	0.96 0.44 0.06	0.70 0.32 0.00
Mixed	Scale	0.392	0.366	0.94 0.41 0.06	0.69 0.79 0.07
Mixed	Shape	0.047	0.047	0.94 0.45 0.05	0.93 0.48 0.07

Table 8: A_n: rejection rates by m ($n = \ell_j = 2m$, $J = 3$, $\theta = 0.5$); null rows 2000 data sets, alternatives 1000.

Setting	m	CUR	ALL	WU-R	WU-F	SW-R	SW-F	SW-PB	G-SW
AlignedNull	20	**0.062**$^{\uparrow}$	0.051	0.057	**0.106**$^{\uparrow}$	0.057	**0.079**$^{\uparrow}$	**0.093**$^{\uparrow}$	**0.060**$^{\uparrow}$
	40	0.051	0.049	0.052	**0.103**$^{\uparrow}$	0.054	**0.068**$^{\uparrow}$	0.029	0.050
	80	0.056	0.052	0.059	**0.110**$^{\uparrow}$	0.052	0.052	0.026	0.056
	160	0.059	0.057	0.055	**0.114**$^{\uparrow}$	0.057	0.057	0.025	0.057
ShiftedNull	20	0.045	0.052	0.046	**0.072**$^{\uparrow}$	0.045	0.053	**0.065**$^{\uparrow}$	0.044
	40	**0.060**$^{\uparrow}$	0.053	0.057	**0.081**$^{\uparrow}$	0.054	**0.082**$^{\uparrow}$	**0.132**$^{\uparrow}$	0.056
	80	0.054	0.049	0.050	**0.068**$^{\uparrow}$	0.049	**0.070**$^{\uparrow}$	**0.114**$^{\uparrow}$	0.050
	160	0.046	0.040	0.047	0.054	0.040	0.041	**0.070**$^{\uparrow}$	0.043
AlignedMean	20	0.223	0.314	0.245	0.375	0.209	0.300	0.339	0.217
	40	0.444	0.569	0.507	0.627	0.494	0.578	0.594	0.462
	80	0.702	0.821	0.775	0.834	0.820	0.821	0.846	0.750
	160	0.957	0.985	0.978	0.985	0.985	0.985	0.983	0.972
BlindSpot	20	0.361	0.060	0.296	0.344	0.375	0.371	0.371	0.366
	40	0.626	0.047	0.526	0.557	0.547	0.564	0.488	0.596
	80	0.890	0.022	0.828	0.855	0.189	0.228	0.147	0.860
	160	0.994	0.007	0.978	0.990	0.009	0.009	0.001	0.992